

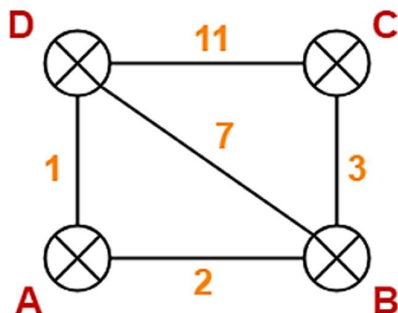
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1. (a) A bit stuffing based framing protocol uses an 8-bit delimiter pattern 01111110. If the output stream after bit stuffing is 01111100101. Then find the original input frame. **(2 M)**
 (b) How pure Aloha utilize the benefits of exponential Back-off Time (T_B) to mitigate the chances of data collision in every attempt of data transmission (K)? Explain with appropriate flow-chart/algorithm. Assume maximum attempt for data transmission (K_{max}) is 15. **(8 M)**

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2. Station A uses 32-Byte frame to transmit messages to Station B using a Selective Repeat protocol. The round-trip delay between A and B is 60 milliseconds, and the bottleneck bandwidth on the path between A and B is 256 kbps.
 (i) What is the optimal window size for Station A to use to achieve maximum efficiency of data utilization? **(5 M)**
 (ii) Determine the minimum number of Sequence Numbers (SN) required and the number of bits required for each SN. **(5 M)**

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3. Assume, sender sends a 4-bit frame (1101) to receiver through a non-reliable channel. Generate the code-word that is being transmitted to receiver side using 7-bit hamming code method. Find out the hamming code received by the receiver if there is no error in the code.
 Now, assume 3rd bit from the left in the transmitted codeword is corrupted during transmission. How receiver will correct the bit using hamming code method. Explain both the scenarios with appropriate steps. **(10 M)**

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4. Suppose a network has an IP address 192.168.1.0. Due to internet security, network is divided into 4 sub-networks. Find the below IP addresses associated with each subnet. **(10 M)**
 (i) Subnet ID, (ii) First configurable Host ID, (iii) Last configurable Host ID, (iv) Direct Broadcast address, (v) Number of configurable hosts

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5. Find the shortest path from each router to all other routers for below mentioned graph by applying Distance Vector Routing scheme. Construct the updated routing tables for all routers. **(10 M)**



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6. A sender wants to transmit the bit sequence 1101011011 using CRC with the generator polynomial $G(x) = x^4 + x + 1$. Corrupt the 4th bit from the right in the transmitted message and show that error is detected by the receiver using CRC technique. **(10 M)**
 - a. Explain working of CRC (Pseudo code)
 - b. Step-by-step workflow of CRC at sender's side
 - c. Write the transmitted codeword after bit corruption
 - d. Compute CRC at receiver's side clearly

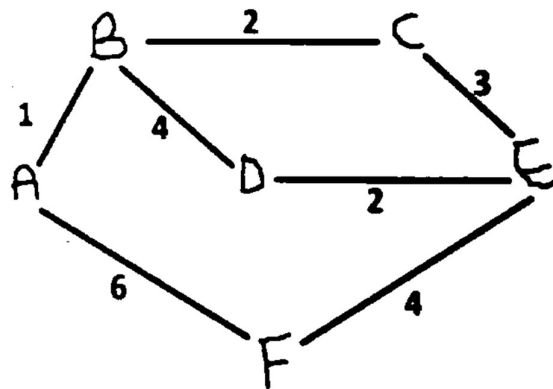
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7. Imagine a situation where there are 12 frames and window size is 4.
 - a. Describe how the sender responds in Go-Back-N protocol after failure of 3rd ACK and 6th ACK **(5 M)**
 - b. Describe how the sender responds in Selective-Repeat protocol after getting NACK for 2nd and 5th frame. **(5 M)**

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8. Compare fixed-size framing and variable-size framing. If Data $\rightarrow$ 011101100110 and ED $\rightarrow$ 0111 then, find data after bit stuffing. Do you consider bit stuffing better than byte stuffing? Justify your statement properly. (10 M)

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9. A university has the IP block 10.0.0.0/24. Implement Variable Length Subnet Masking (VLSM) to allocate IP addresses to the following subnets. List the following for each subnet: Subnet ID, Broadcast Address, Range of Addresses, and Subnet Masks. (10 M)

- a. Subnet A: 100 hosts
- b. Subnet B: 50 hosts
- c. Subnet C: 25 hosts
- d. Subnet D: 10 hosts

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10. A router in a link state routing network receives link state advertisements (LSAs) from other routers and builds the following topology with link costs shown in the diagram. You are the network engineer for router A. Use Dijkstra's algorithm to compute the shortest path from A to all other remaining routers. Show all the steps involved and create the final routing table at router A. (10 M)



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11. A 4-bit data word 1011 is to be transmitted using Hamming (7,4) code with even parity. Encode the data word into a 7-bit Hamming code, showing the placement and calculation of the parity bits. Suppose the received codeword is 0110111; detect whether there is an error, locate the erroneous bit if any, correct it, and recover the original 4-bit data word. (10 M)

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12. A group of N stations share a 200 Kbps slotted ALOHA channel. Each station generates a 1000-bit frame on average every 10 seconds, regardless of whether the previous frame was transmitted. What is the maximum number of stations N that can share the channel efficiently (i.e., so that the offered load $G = 1$)? (10 M)

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13. A 20 Kbps satellite link has a one-way propagation delay of 400 ms. The transmitter uses Go-Back-N ARQ with a window size of $N = 10$, and each frame contains 100 bytes of user data. Compute the transmission time for one frame, and assuming ideal acknowledgments (negligible

size and processing) with the sender continuously filling its window, determine the maximum achievable throughput (in kbps) of user data delivered to the receiver. **(10 M)**

14. An ISP assigns the network 210.10.0.0 to a company. **(10 M)**

4 (i) Under classful addressing, determine the class of the address, its default mask, and how many usable hosts it supports.

(ii) Using classless addressing, the company wants 8 equal subnets.

- What is the new subnet mask (prefix length)?
- How many usable hosts are there per subnet?
- Write the network address, first usable IP, and broadcast address of the first subnet.

15. Consider the following data and design a network. Use Distance Vector Routing Algorithm to find the shortest path from each node to all other nodes in the network. **(10 M)**

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Nodes and Distances	A	B	C	D
A	0	3	5	1
B	3	0	4	∞
C	5	4	0	2
D	1	∞	2	0

16. Assume we have a 7-bit data sequence that needs to be transmitted using a Hamming code with even parity. The data sequence to be transmitted is 1011010. Calculate the required number of parity bits and their positions, generate the Hamming code by adding the parity bits to the data sequence, simulate an error in the MSB bit in the received code and detect the error using the Hamming code technique. **(10 M)**

- 3 17. Compare Pure ALOHA and SLOTTED ALOHA. Consider a group of N stations share 100 Kbps slotted ALOHA channel. Each station outputs a 100-bit frame on an average of 1000 ms even if previous one has not been sent. What is the required value of N? **(10 M)**

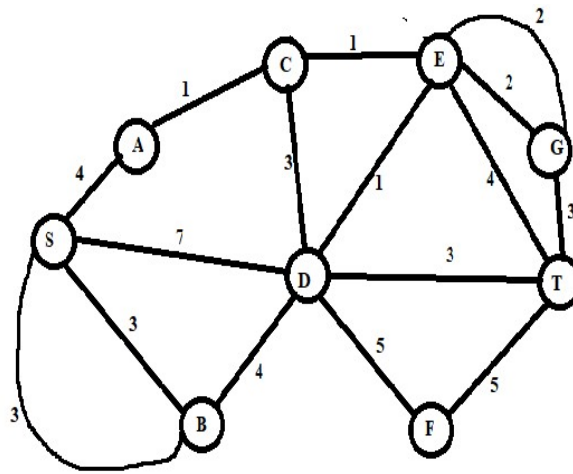
- 3 18. A 3000 km long trunk operates at 1.536 Mbps and is used to transmit 64-byte frames. The link uses a sliding window protocol, and the propagation delay is 6 μ s per km. Determine the number of bits required for the sequence number field to fully utilize the link. **(10 M)**

- 4 19. To distribute the block of addresses (170.100.0.0/16) to the three groups of customers as specified, we need to calculate how many addresses each group requires and then determine how to subnet the address block accordingly. **(10 M)**

The requirements for each group:

- First group: 10 customers * 256 addresses/customer = 2560 addresses
- Second group: 6 customers * 128 addresses/customer = 768 addresses
- Third group: 2 customers * 64 addresses/customer = 128 addresses

20. Find the least cost route between the node S to F for the given network using Link State Routing. Show all the steps involved in finding the least cost route. **(10 M)**



21. Assume that, in a Stop-and-Wait ARQ system, the bandwidth of the line is 2 Mbps, and 1 bit takes 25 ms to make a round trip. What is the bandwidth-delay product? If the system data frames are 1200 bits in length, what is the utilization percentage of the link? **(10 M)**
22. Encode the 4-bit data word 1010 using Hamming Code. Show the complete sender-side (encoding) process and then demonstrate how the receiver detects and corrects a single-bit error (3rd bit from the left in the hamming code) in the received codeword. **(10 M)**
23. A group of M stations share a 200 Kbps slotted ALOHA channel. Each station outputs a 1000-bit frame on an average of 2000 ms even if the previous one has not been sent. What is the required value of M? **(10 M)**
24. We have a big network having IP Address 192.68.223.186/20. We want to divide this network into 4 subnets. Complete the process and answer the following questions. **(10 M)**
 - Range of IP addresses in each subnet
 - First usable address and last usable address in each subnet
 - Network address of each subnet
 - Broadcast address of each subnet
 - Subnet Mask of each subnet
25. Consider the following data and design a network. Use Link State Routing Algorithm to find the shortest path from each node to all other nodes in the network. **(10 M)**

Nodes and Distances	A	B	C	D
A	0	1	3	1
B	1	0	4	∞
C	3	4	0	2
D	1	∞	2	0

26. The following character encoding is used in a data link protocol:
A: 01000111; B: 11100011; C: 10101111; FLAG: 01111110; ESC: 11100000
Show the bit sequence transmitted (in binary) for the six-character frame when byte stuffing is used.
A ESC B FLAG C ESC
Also explain the difference between fixed size framing and variable size framing in detail. **(10 M)**

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27. A 7-bit message 1001100 is to be transmitted using Hamming codes with even parity. Construct the codeword. If a single-bit error occurs at 4th position from the left during transmission, show how the receiver detects and corrects it. **(10 M)**

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28. a) A network using CDMA/CD has a bandwidth of 10 Mbps. The maximum propagation time including the delays in the devices and ignoring the time needed to send a jamming signal is 25.6 μ sec. What is the minimum size of the frame? **(4 M)**

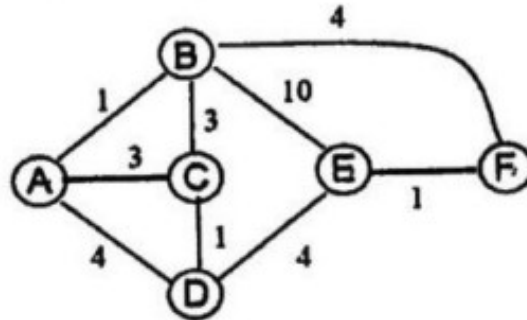
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- b) A sender and receiver are communicating using the Go-Back-N protocol with a window size of 4 and sequence numbers 0 to 7. The sender transmits frames 0, 1, 2, 3, 4, 5, 6, but frame 3 is lost during transmission while frames 4, 5, 6 arrive at the receiver out of order. Explain how the receiver handles the out-of-order frames, how the sender retransmits frames after a timeout, how the sliding window moves at both ends. **(6 M)**

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29. TechSolutions has been assigned the network 192.168.10.0/24 for its four departments: HR, Sales, IT, and Admin. Identify the network class and divide the network into four subnets. Find out the subnet ID, broadcast address, first usable address, last usable address and subnet mask for each subnet. Discuss the process in detail. **(10 M)**

30. Consider the network shown below. Show the operations of Link State (Dijkstra's) algorithm for computing the least cost path from router F to all destinations. Explain the process. **(10 M)**



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